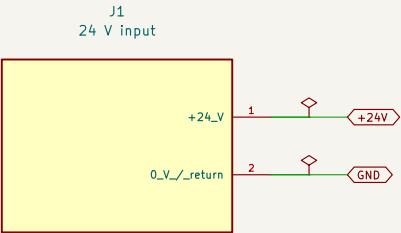
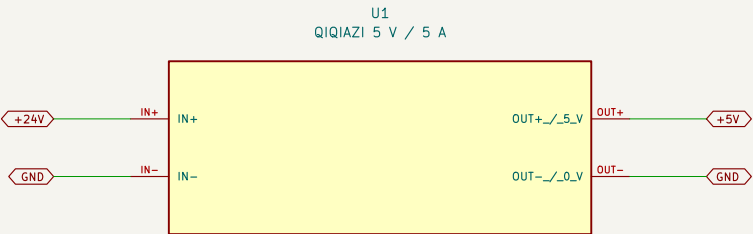


CONTROLLER AND POWER

A



B



Wire converter by IN+/IN-/OUT+/OUT- markings.
24 V also feeds all three motor drivers.
Common 0 V; separate power branches to each driver.
5 A is the converter rating, not the Nucleo input rating.

C

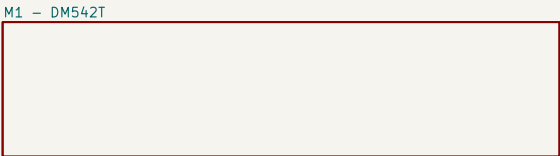
NUCLEO POWER CONFIGURATION
JP2: fit EXT jumper on pins 5–6.
Buck +5 V -> CN11 pin 6; ground -> CN11 pin 8.
CN8 pin 7 supplies 3.3 V to interface/storage.
5V_EXT input: 4.75–5.25 V, 500 mA maximum.
Power EXT first, then connect ST–Link USB CN1.
Do not connect 24 V to Nucleo VIN or logic pins.

D

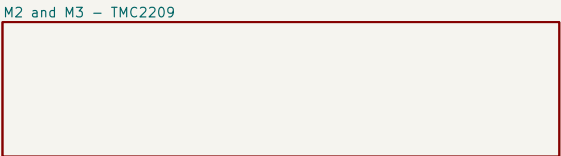
BOARD ROUTING (UM2407 table 14)
PE9 -> CN10.4; SB28 ON, SB70 OFF.
QSPI: SB59/SB63 ON; SB61/SB69 OFF.
PE2 -> CN10.25; SB67 OFF.
Only used external contacts are drawn.
Ethernet, ST–Link, LEDs and button stay onboard.

Shared names connect across sheets. A1 contacts use CNxx.pin numbering. J101–J111 are perfboard headers.
REF inputs: internal 3.3 V pull-ups, both-edge EXTI; NC to GND. Released = LOW; pressed / unplugged = HIGH.
External modules and switches are excluded from board; interface parts have linked footprints.

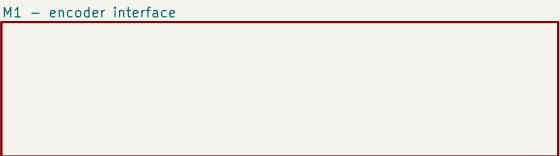
E



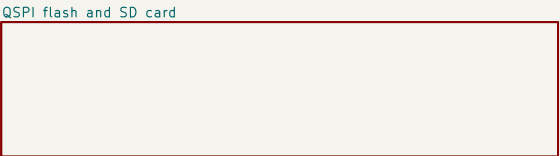
Date: sheets/M1_DM542T.kicad_sch



Date: sheets/M2_M3_TMC2209.kicad_sch

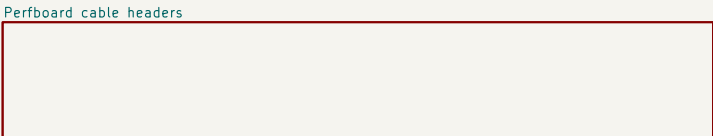


Date: sheets/M1_Encoder.kicad_sch

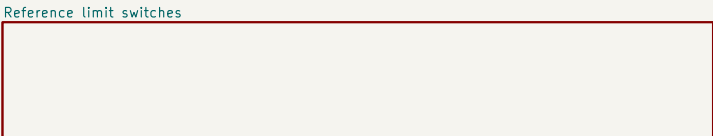


Date: sheets/Storage.kicad_sch

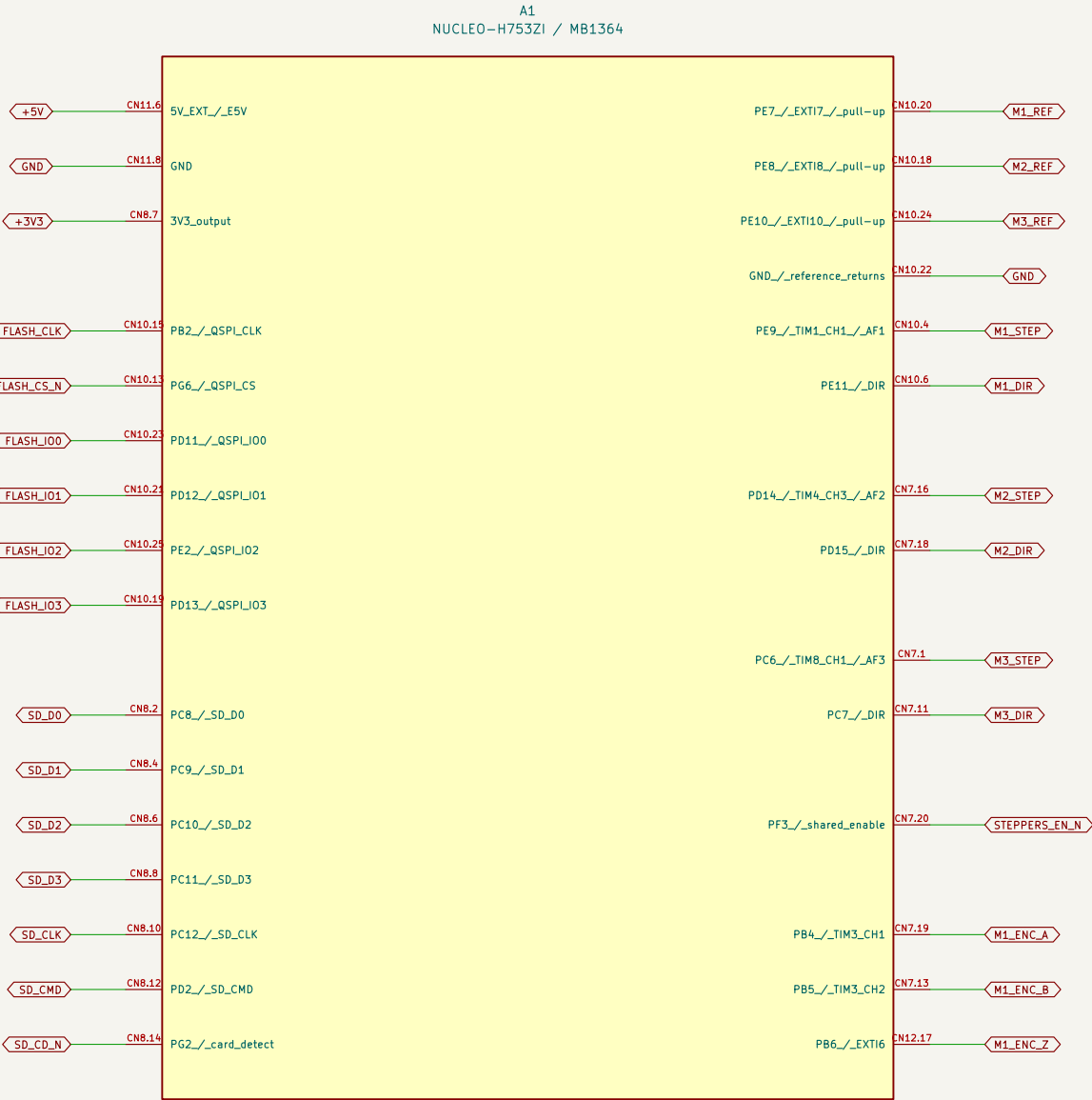
F



Date: sheets/Perfboard_Connectors.kicad_sch



Date: sheets/Reference_Switches.kicad_sch



A

B

C

D

E

F

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /
File: ProtoMate.kicad_sch

Title: ProtoMate SCARA – Controller and power

Size: A3 Date: 2026–10–03

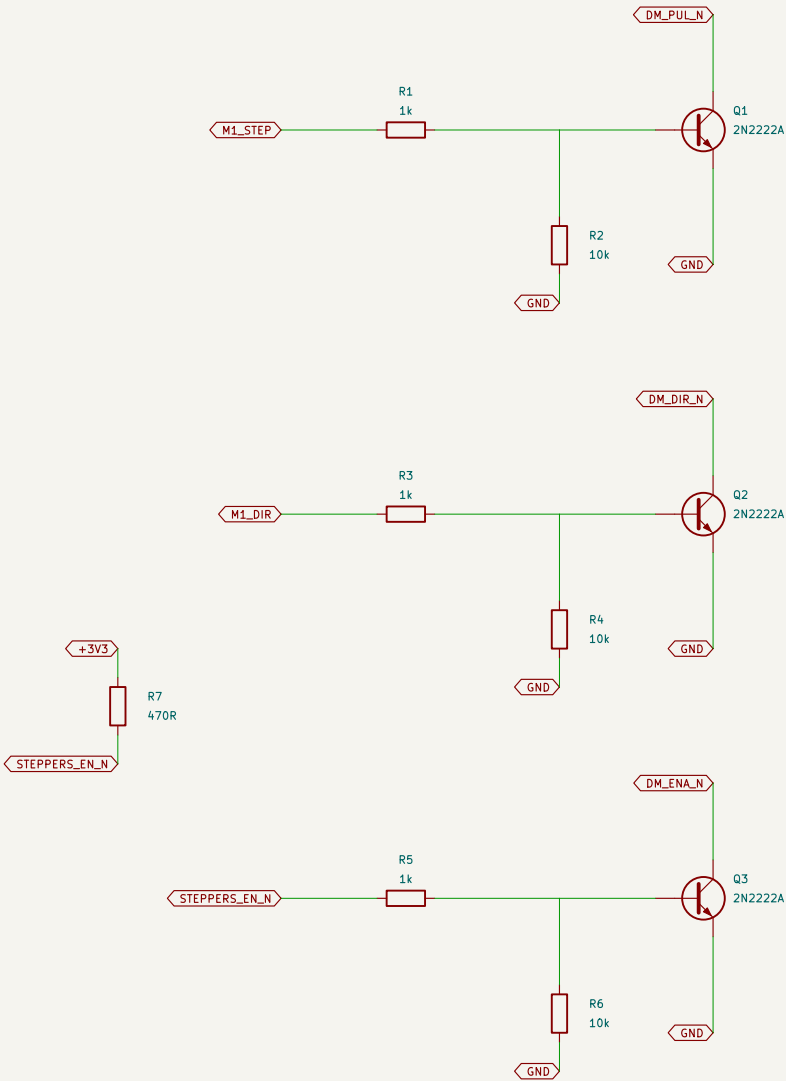
KiCad E.D.A. 10.0.6

Rev: 4.0

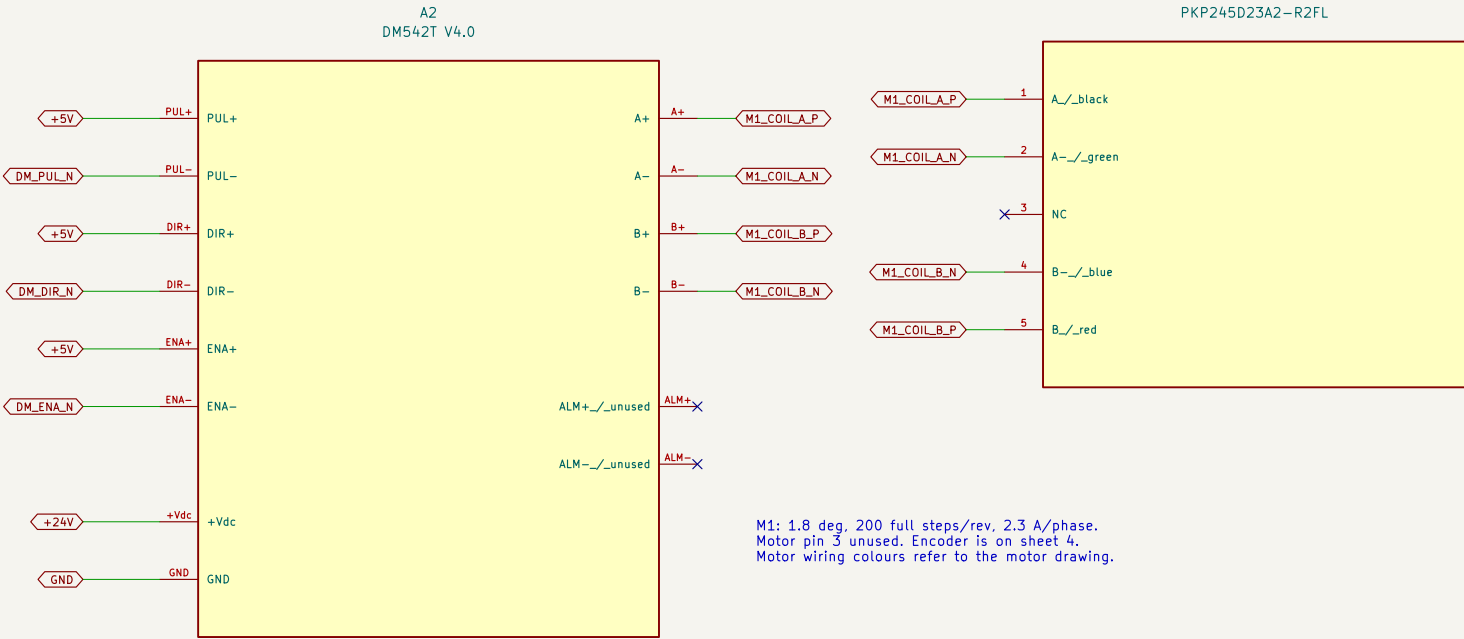
Id: 1/7

M1 – DM542T

3.3 V MCU -> 2N2222A current sinks -> 5 V optoisolated inputs



SHARED ENABLE AND TIMING
PF3 low: Q3 off, DM542T enabled, both TMC EN inputs low.
PF3 high: Q3 on, DM542T disabled, both TMC EN inputs high.
R7 pulls PF3 high during MCU reset; about 7 mA sink when enabled.
After enable, wait >=200 ms. DIR setup >=5 us; STEP high/low >=2.5 us.



DM542T SETTINGS / INITIAL COMMISSIONING
S2 signal selector: 5 V.
1/16 step: SW5 ON, SW6 ON, SW7 OFF, SW8 ON.
3200 STEP pulses per motor revolution.
Conservative initial current: 2.03 A RMS / 2.84 A peak
SW1 ON, SW2 ON, SW3 OFF; SW4 OFF (idle reduction).
Check the actual V4.0 label before setting switches.

Q1–Q3 show functional B/C/E pins, not package lead numbers.
Verify your 2N2222A manufacturer/package before assembly.
Fit each 10k between base and emitter. Keep PUL+ /PUL- paired;
separate signal cables from motor power cables.

System wiring and linked perfbboard assembly map | all voltages DC

ProtoMate Control		
Sheet: /M1 – DM542T/ File: M1_DM542T.kicad_sch		
Title: ProtoMate SCARA – M1 – DM542T		
Size: A3	Date: 2026–10–02	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 2/7

M2 AND M3 – TMC2209

MS1 = MS2 = VDD (3.3 V): 1/16 step, 3200 pulses/rev. No MCU pins needed.
SPRD jumper open: factory StealthChop mode. Current set with each onboard potentiometer.

A

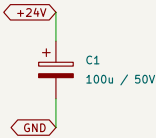
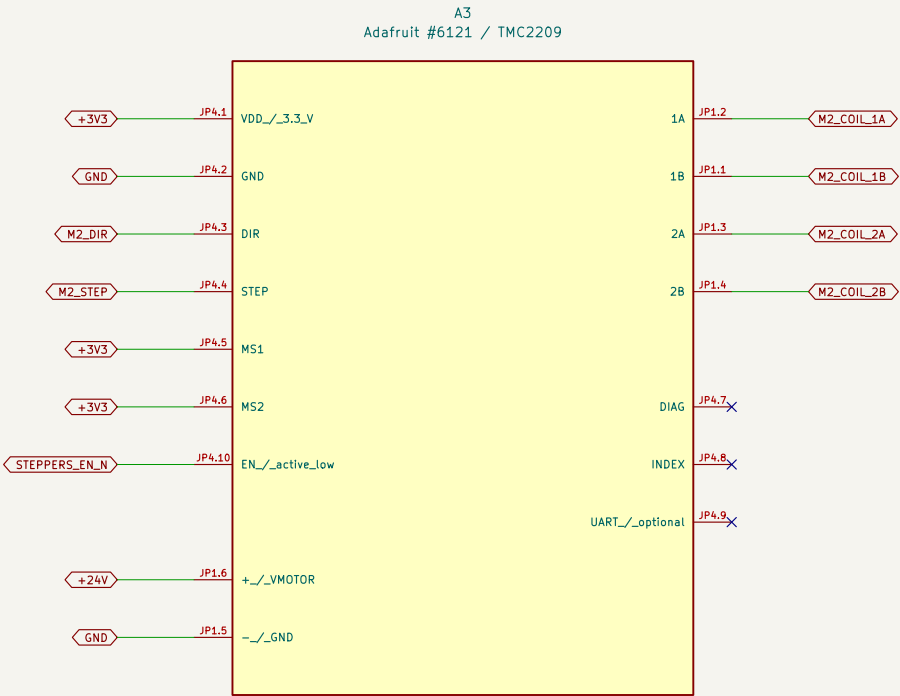
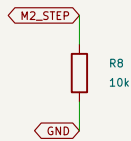
B

C

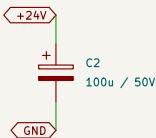
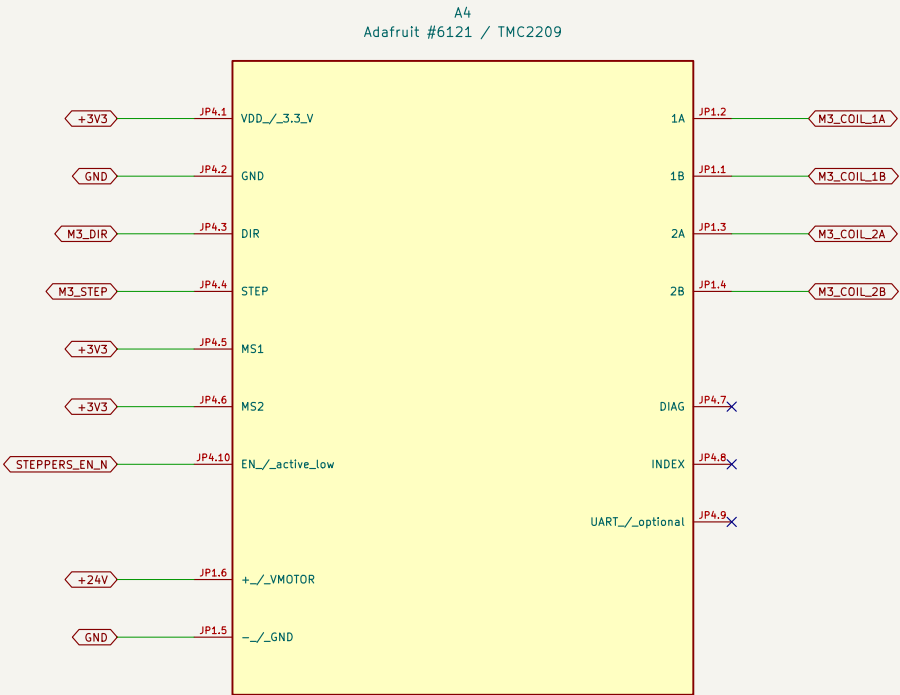
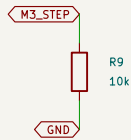
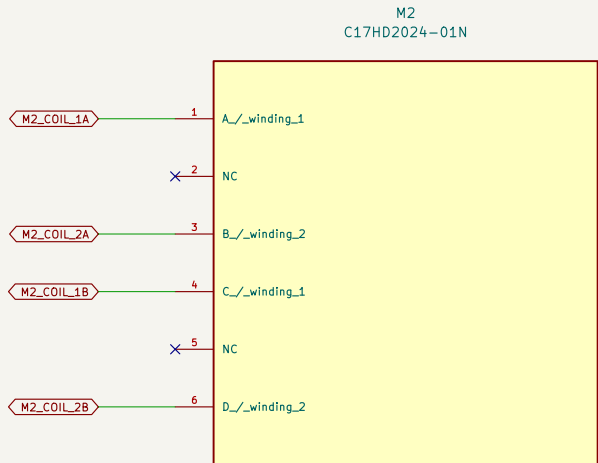
D

E

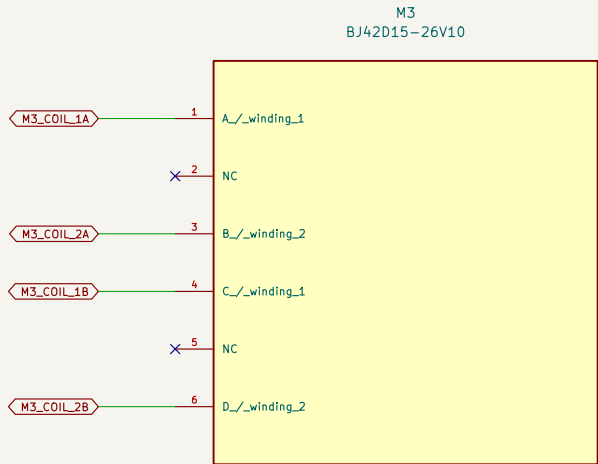
F



Extra bulk capacitor near VMOTOR.
Onboard bypass/bulk already included.
Keep VMOTOR below 29 V including
deceleration transients.



Extra bulk capacitor near VMOTOR.
Onboard bypass/bulk already included.
Keep VMOTOR below 29 V including
deceleration transients.



M2: 1.0 A/phase, 4.1 ohm, 8.5 mH; coil pairs 1–4 and 3–6 (user-confirmed).
M3: 0.84 A/phase, 6.0 ohm, 8.8 mH: 20T / 2 mm pulley -> 40 mm/rev -> 80 pulses/mm.
Both motors: 1.8 deg. Adjust current separately: respect peak/RMS conventions.
JP1/JP4 identifiers are Adafruit board contacts; follow the printed terminal labels.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /M2 and M3 – TMC2209/

File: M2_M3_TMC2209.kicad_sch

Title: ProtoMate SCARA – M2 and M3 – TMC2209

Size: A3

Date: 2026-10-02

Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 3/7

A

B

C

D

E

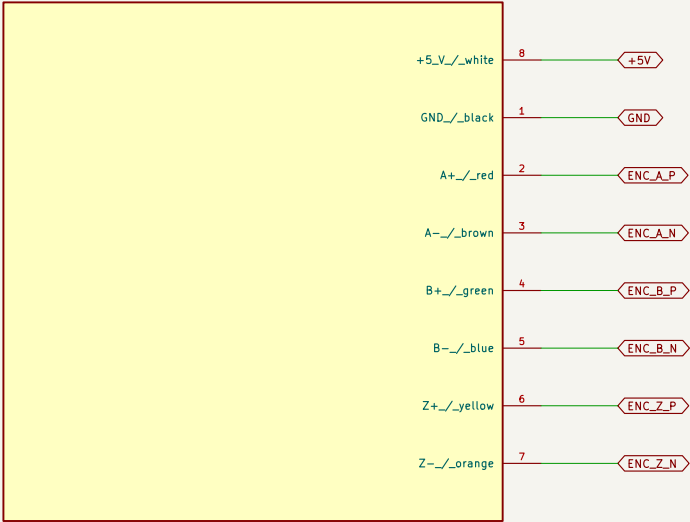
F

M1 – ENCODER / HC126 + NPN LEVEL SHIFTERS

U2 input pairs are deliberately reversed.
Q4–Q6 cancel that inversion before U3.
Encoder connector pinout stays unchanged.
Never wire U2 outputs directly to U3.

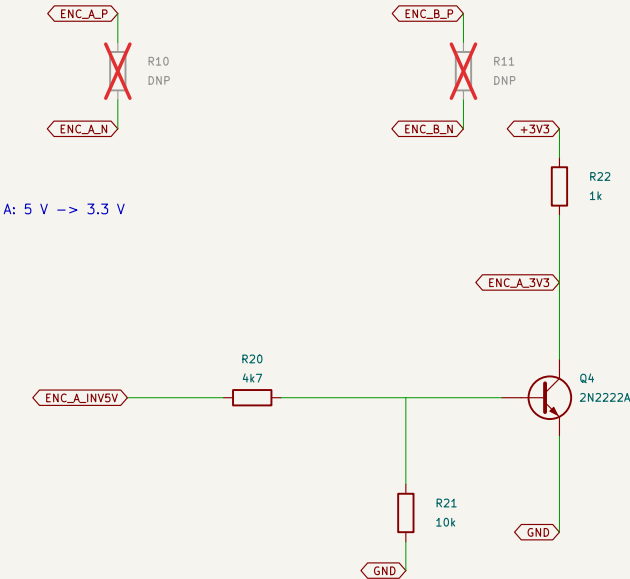
U2: 5 V. Q4–Q6 translate to 3.3 V. U3: 3.3 V ONLY.
U3 OE1/OE2/OE3 HIGH; unused OE4 and A4 LOW.
U2 unused receiver inputs may float (fail–safe).
100 nF at each IC; 1 uF nearby.

J2
M1 encoder / 8–pin

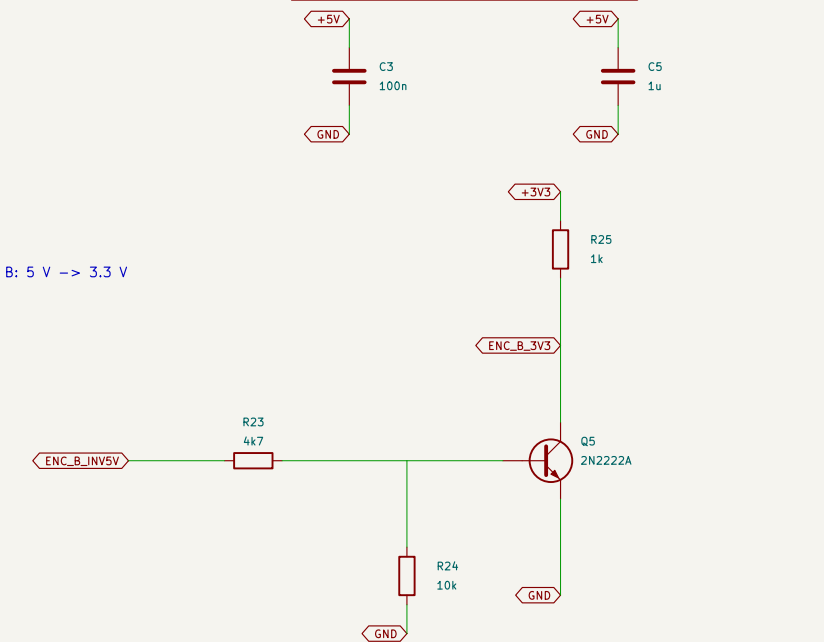


400 P/R -> TIM3 x4 = 1600 counts/rev; Z = once/rev.
Cable colours / connector orientation; motor manual.
Shielded twisted pairs; chassis bond at cable entry.

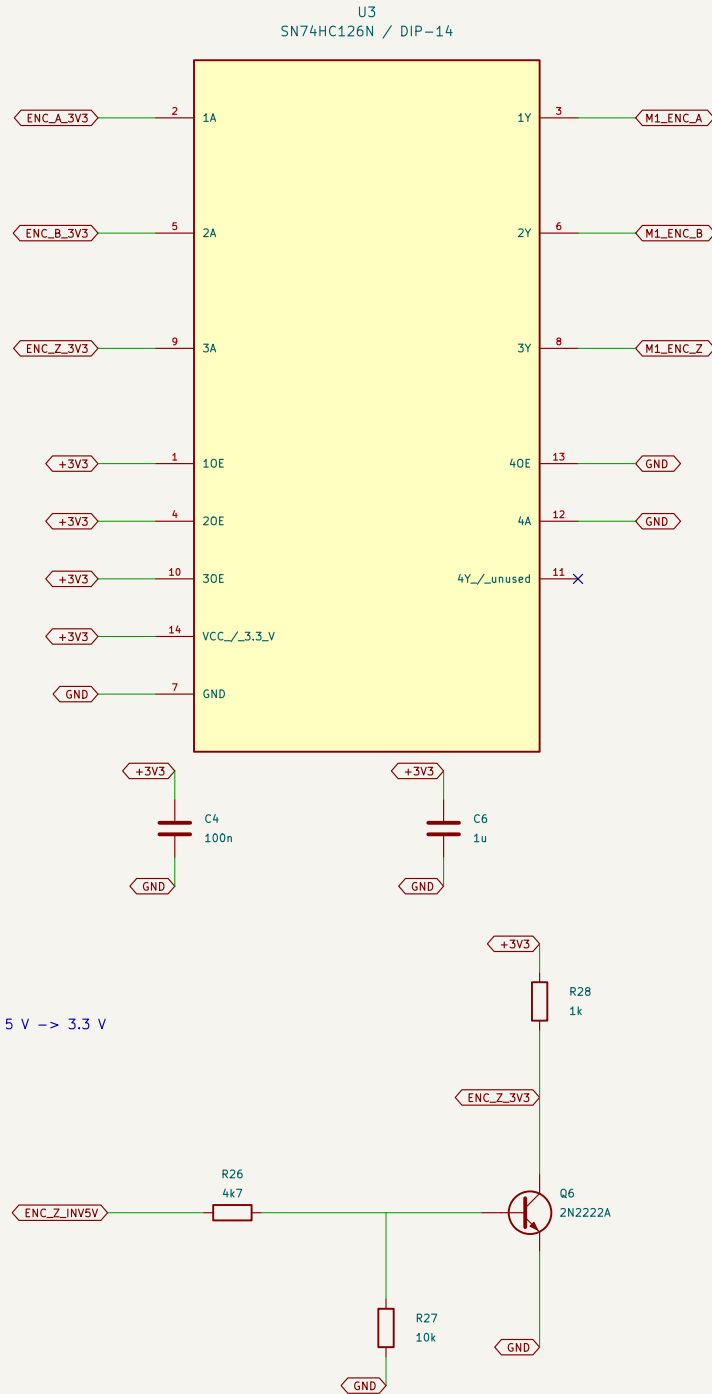
R10–R12: OPTIONAL TERMINATION, DNP.
Choose for cable and 20 mA encoder output limit.
A+/A-, B+/B-, Z+/Z-: twisted pairs.



A: 5 V -> 3.3 V



B: 5 V -> 3.3 V



Z: 5 V -> 3.3 V

A/B -> PB4/PB5; Z -> PB6 rising–edge EXTI. All original signal polarities preserved.
Q4–Q6: Diotec 2N2222A E=1/B=2/C=3. Keep collector–to–U3 wires short.
Check edge shape and transistor turn–off delay at the intended encoder speed.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /M1 – encoder interface/
File: M1_Encoder.kicad_sch

Title: ProtoMate SCARA – M1 – encoder interface

Size: A3

Date: 2026–10–03

Rev: 2.0

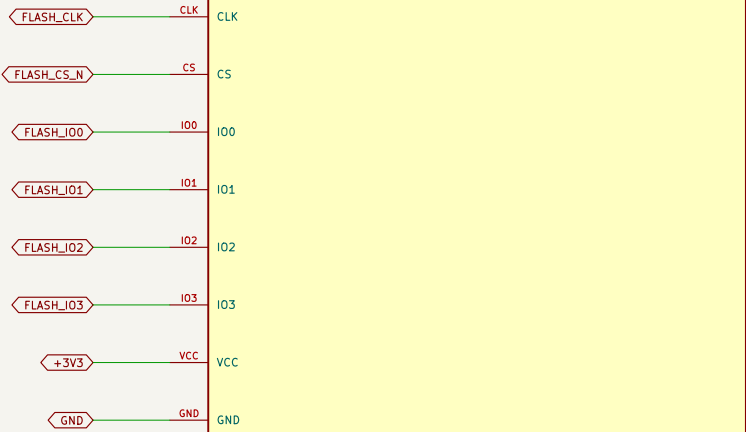
KiCad E.D.A. 10.0.6

Id: 4/7

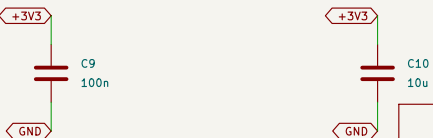
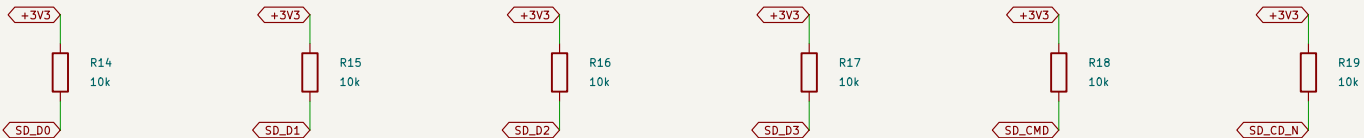
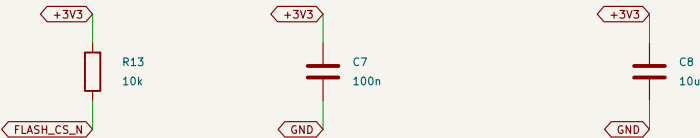
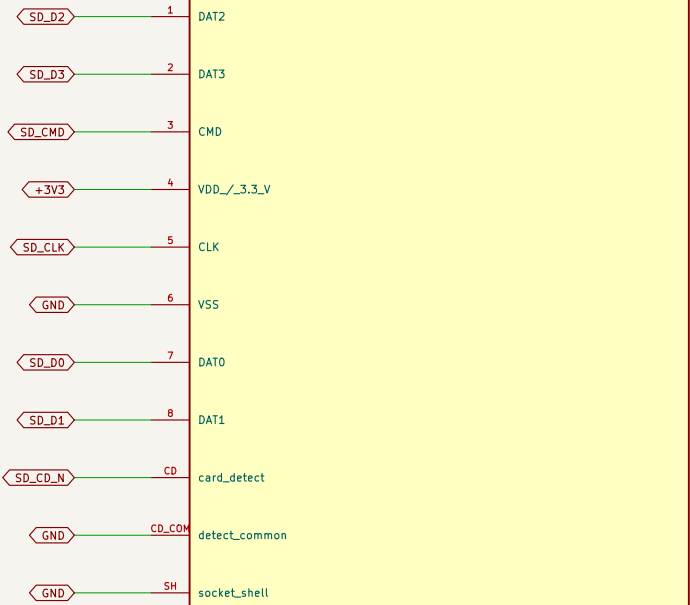
QSPI FLASH AND SD CARD

SD pins 1–8 use standard card–contact numbering.
CD switch closes to GND when inserted (advisory input).
Pull–ups on CMD/D0–D3/CD; no pull–up on CLK.
Existing equivalent module pull–ups/caps need not be duplicated.

A5
W25Q128 / 3.3 V QSPI



J3
microSD / 4–bit socket module



3.3 V ONLY
Use a W25Q128 module exposing IO0–IO3.
IO2/WP and IO3/HOLD are data in quad mode;
do not hard–wire them to supply or ground.
Module contacts are identified by signal name.
C7/C8 and R13 may already exist on the module;
fit them if absent. Keep the QSPI harness short.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /QSPI flash and SD card/

File: Storage.kicad_sch

Title: ProtoMate SCARA – QSPI flash and SD card

Size: A3

Date: 2026–10–02

Rev: 1.0

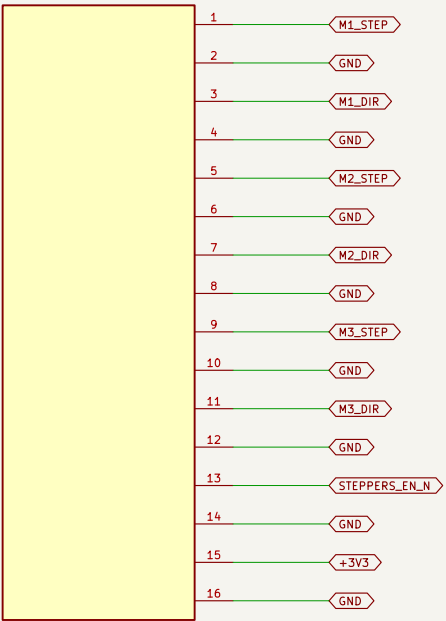
KiCad E.D.A. 10.0.6

Id: 5/7

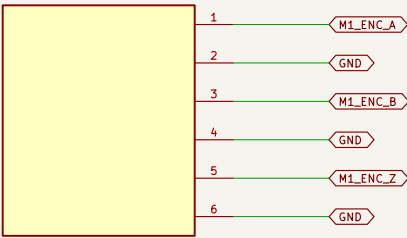
PERFBOARD CABLE HEADERS

Pin numbers match the layout and Harness.csv. All headers are 2.54 mm pitch; every drawn contact is on the perfboard.

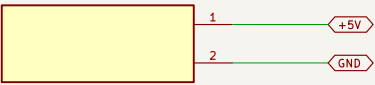
J101
Nucleo outputs / 2x08



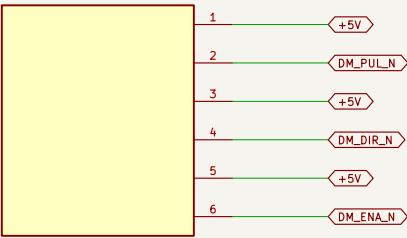
J102
Nucleo encoder / 2x03



J103
Buck 5V / 1x02



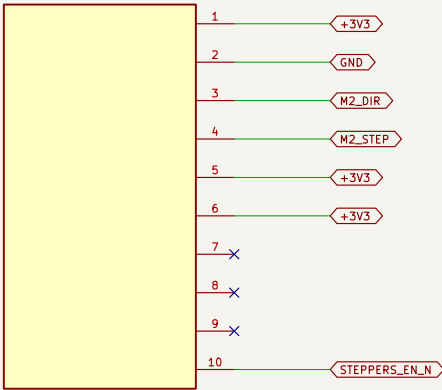
J104
DM542T / 1x06



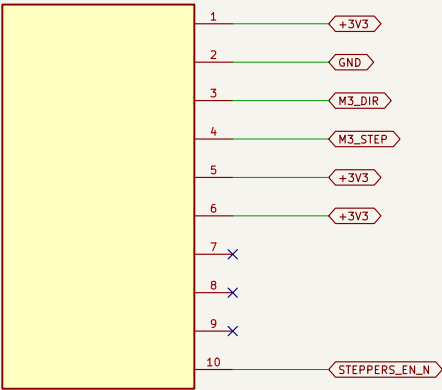
J101/J102: paired even pins are GND.
J101 pin 15 receives Nucleo 3.3 V.
J103: regulated 5 V from buck only.
24 V and motor currents stay off this board.

J104: 1 PUL+, 2 PUL-, 3 DIR+, 4 DIR-, 5 ENA+, 6 ENA-.
J105/J106: 1 VDD, 2 GND, 3 DIR, 4 STEP, 5 MS1, 6 MS2,
7 DIAG (NC), 8 INDEX (NC), 9 UART (NC), 10 EN.
J107 follows the numbered J2 encoder connector.

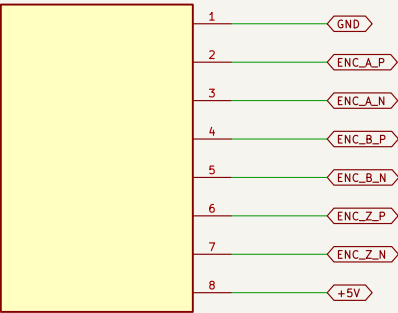
J105
M2 TMC2209 / 1x10



J106
M3 TMC2209 / 1x10



J107
M1 encoder / 1x08



Read Harness.csv for destination contacts (CNxx.pin / driver labels).
External module symbols elsewhere show the far end of these cables. Matching names represent the same conductor.
J108–J111 (Nucleo reference inputs and the three NC switches): see the Reference limit switches sheet.

All J101–J107 headers mount on the perfboard; signals connect by global names

ProtoMate Control

Sheet: /Perfboard cable headers/

File: Perfboard_Connectors.kicad_sch

Title: ProtoMate SCARA – Perfboard cable headers

Size: A3

Date: 2026–10–03

Rev: 3.0

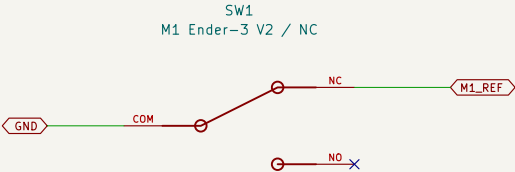
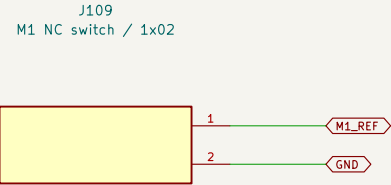
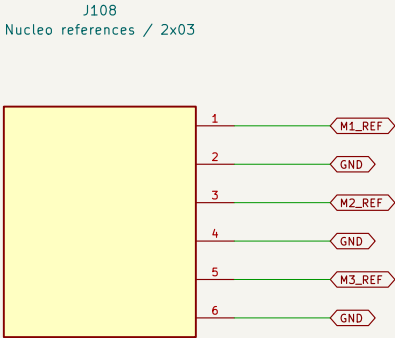
KiCad E.D.A. 10.0.6

Id: 6/7

REFERENCE LIMIT SWITCHES – NORMALLY CLOSED

J108–J111 mount on the perfboard. SW1–SW3 mount on the robot. Global labels connect the cable endpoints.

Switches shown released (COM–NC closed).



J108 -> Nucleo A1
1 = PE7 / CN10.20 (M1)
3 = PE8 / CN10.18 (M2)
5 = PE10 / CN10.24 (M3)
2, 4, 6 = GND / CN10.22

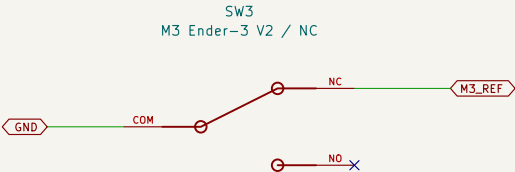
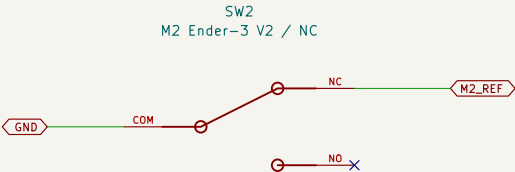
INPUT BEHAVIOR
Nucleo internal pull–ups to 3.3 V.
Released: COM–NC closed -> LOW.
Pressed / unplugged: open -> HIGH.
Both–edge EXTI: raw, un–debounced inputs.
Homing and motor–stop logic not yet implemented.

BEFORE WIRING: With the switch disconnected, verify continuity: closed when released, open when pressed.
COM / NC / NO identify contact functions, NOT the module connector pin order. Do not infer power wiring from V/G/S printing.
Use the unpopulated mechanical version as a dry contact; no supply connection. Route each REF with its GND return, away from motor wiring.

J109.1 -> verified NC contact
J109.2 -> verified COM contact
NO contact stays unconnected.

J110.1 -> verified NC contact
J110.2 -> verified COM contact
NO contact stays unconnected.

J111.1 -> verified NC contact
J111.2 -> verified COM contact
NO contact stays unconnected.



NC dry contacts | PE7 / PE8 / PE10 | 3.3 V internal pull–ups

ProtoMate Control

Sheet: /Reference limit switches/
File: Reference_Switches.kicad_sch

Title: ProtoMate SCARA – Reference limit switches

Size: A3 Date: 2026–10–03 Rev: 4.0
KiCad E.D.A. 10.0.6 Id: 7/7